

The summary of the diploma project of the student: MOREANU P.D. Vlad-Ştefan, group 441G

Study program: Telecommunication systems and technologies

# Three channel stereo audio mixer

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**Project goals:** The purpose of the project is the practical implementation, with discrete components of the audio mixer with output of both the mixed analog audio and channel inputs as digital signals, meant to be transferred via USB to a PC workstation for digital processing. Each channel will have a three band equalisation stage (before mixing) and the analog to digital conversion will respect the CD Audio standard specifications. The project involves two design stages with the goal of measuring the performance parameters of the audio mixer. The first stage consists of a theoretical model, created in the SIMetrix simulation engine. The second stage is the practical implementation of the mixer, requiring the design of a PCB on which the required circuitry will be assembled. For the design, the KiCAD PCB editor is used.

**Project development and results:** A simulation schematic was created in SIMetrix v9.1 according to common mixer architecture and design principles. Given the limitations imposed by the thesis of the project, the model was created while taking into account the non-ideal elements of common design such as noise, supply voltages, limited headroom and input and output impedances.

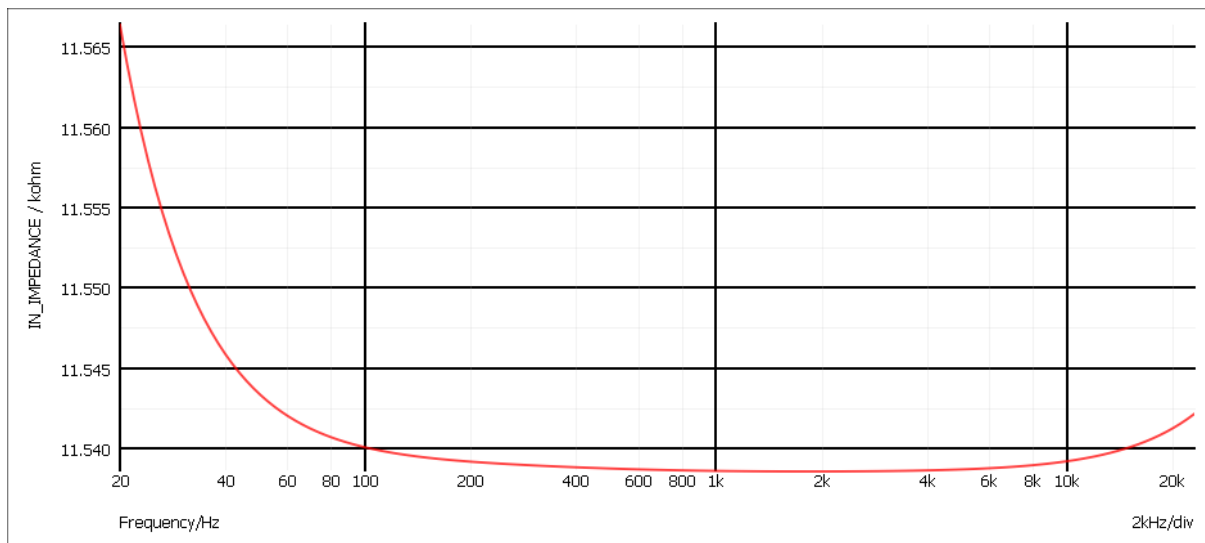


Figure 1: Example of simulation results: The circuit input impedance

After the completion and analysis of the theoretical model, the PCB layout was created in KiCAD v8.0, while trying to minimise effects such as channel crosstalk. The board seen at 3 was created according to the schematic seen at 2.

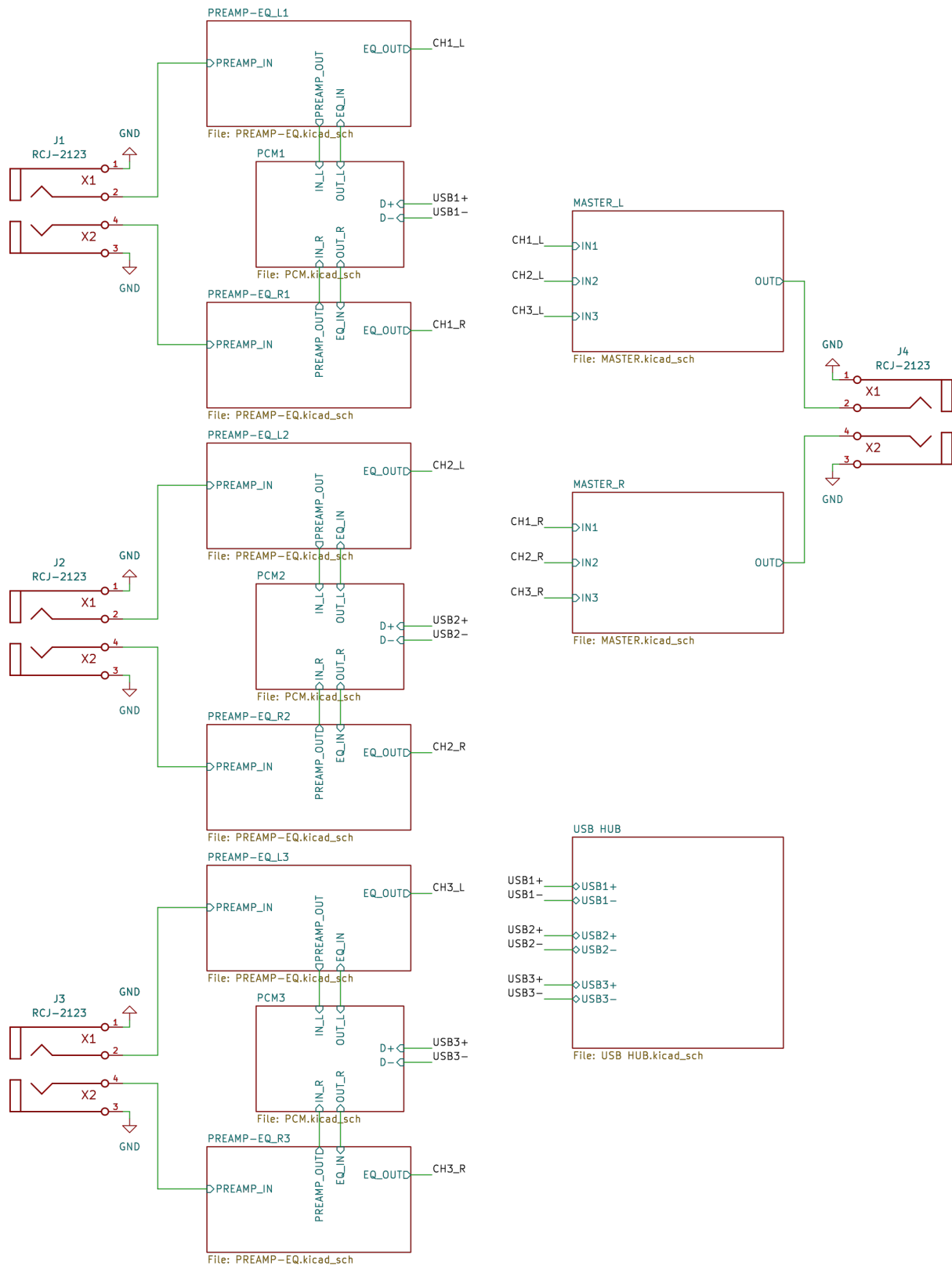


Figure 2: Hierarchic schematic which was implemented

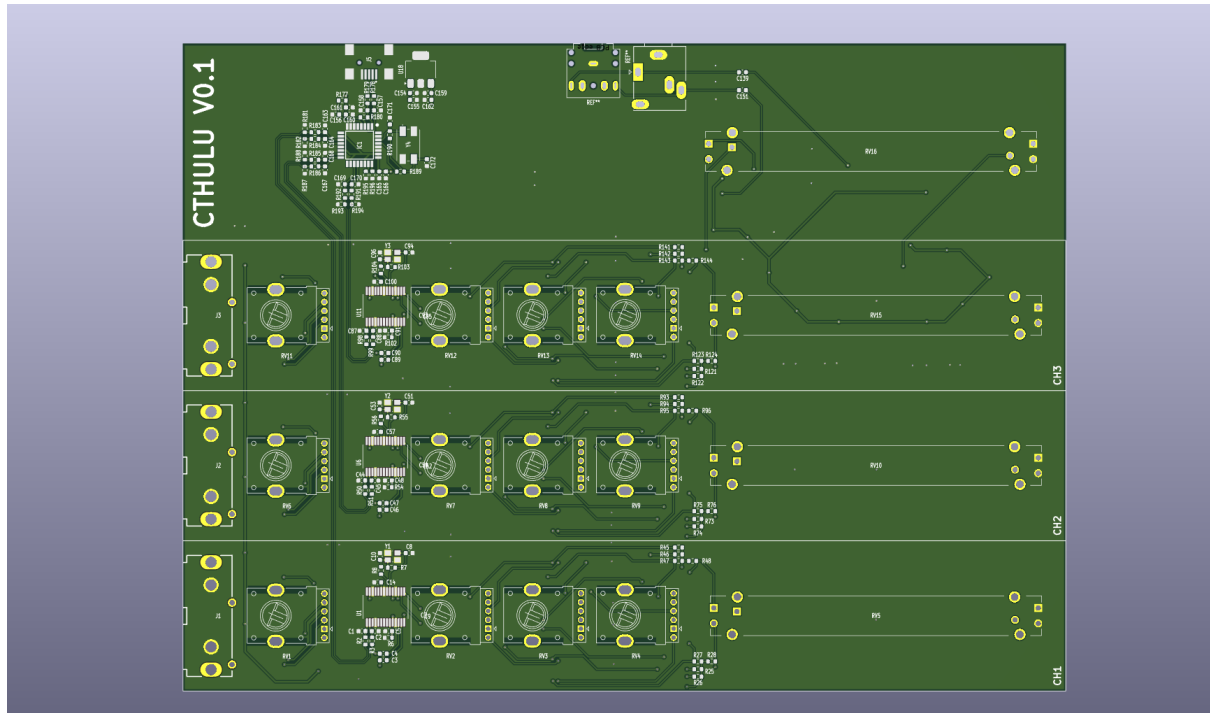


Figure 3: The front side of the PCB